Atharva Nikam

Mechanical Design & Simulation Engineer | CAD · FEM · MATLAB/Simulink · Python

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PROFESSIONAL SUMMARY

Mechatronics Master's student with a mechanical engineering background, combining **MATLAB/Simulink system simulation** and **EV powertrain & battery modeling** with **3D CAD design (SolidWorks, CATIA V5)** and **FEM structural analysis (ANSYS)**. Experienced in taking a design from **concept and simulation** to **prototype, testing and production-ready documentation**, which makes me a strong fit for roles at the interface of **virtual development and hands-on validation**. **Availability:** as soon as possible, **up to 20 h/week** during the semester. **Visa status: valid residence permit** (studies), working-student employment permitted.

EDUCATION

Sep 2024 – Present **M.Sc. in Mechatronics** | *Ravensburg-Weingarten University of Applied Sciences, Germany*
Expected graduation: Aug 2027 | Relevant courses: Advanced Hybrids, Electric Drives, Power Electronics

Dec 2020 – Jul 2023 **B.E. in Mechanical Engineering** | *Bharati Vidyapeeth College of Engineering, India*
Relevant courses: Machine Design, Product Design & Development, Manufacturing Processes

PROJECTS

Sep 2026 – Present **2-Motor BEV Energy Management with MPC** | *MathWorks Capstone Project (#246) – Repo* | MATLAB, Simulink

- Targeting **lower energy losses in a dual-motor EV** by developing a **Model Predictive Control (MPC) strategy** in **MATLAB/Simulink** that distributes torque between front and rear motors.
- Basing every torque-split decision on **motor efficiency maps**, so the controller **balances drive demand against powertrain losses** over the upcoming driving horizon.

Mar 2026 – Jun 2026 **Predictive Electric Vehicle Cooling – Repo** | *MathWorks Capstone Project (#194) – Repo* | MATLAB, Simulink, Simscape Fluids

- Created a **virtual test environment for EV battery temperature** by coupling **vehicle dynamics with a battery model in MATLAB/Simulink** across driving, regenerative braking and DC fast-charging at 25–38 °C ambient.
- Modeled a complete **liquid battery-cooling circuit** (cold plate, pump, radiator) in **Simscape Fluids**, enabling **closed-loop temperature control** of the battery.
- Identified the better cooling strategy by **comparing reactive vs. predictive control** on **temperature-limit violations, cooling energy and fast-charging power**.

Nov 2025 – Jan 2026 **EV Powertrain Simulation & Battery SOC Estimation – Repo** | *Independent Project* | MATLAB

- Achieved **battery state-of-charge (SOC) estimation within 1.29 % error** by implementing a **Kalman filter in MATLAB** and validating it against **standard drive-cycle data**.
- Built an **EV powertrain model** covering battery electrical and thermal behaviour, energy management and regenerative braking to **evaluate system performance under varying operating conditions**.

Nov 2024 – Jan 2025 **Computer Vision: Augmented Reality & Object Detection with Depth Estimation** | *RWU M.Sc. Course Projects (Prof. Dr. Stefan Elser)* | Python, OpenCV, NumPy, YOLOv8

- Projected **virtual posters onto real walls with correct perspective** by detecting **ArUco markers** and applying **homography-based image warping** in **Python (OpenCV, NumPy)**, verified on a given dataset and own recordings.
- Detected **cars in real driving scenes (KITTI dataset) with YOLOv8** and assessed **detection quality via IoU, precision and recall** against ground-truth labels.
- Estimated **distance to detected cars from a single camera image** using the **camera intrinsic matrix and a ground-plane model**, with **estimates closely matching ground truth up to approx. 40 m**.

Jan 2022 – Feb 2023 **Multi-Purpose Agricultural Vehicle** | *Bharati Vidyapeeth College of Engineering* | AutoCAD, SolidWorks

- Reduced **vehicle weight by approx. 12 %** through **structural design improvements** to the chassis, developed together with the fabrication team.
- Confirmed **prototype performance within 5 % of design calculations** through **field testing**.

- Designed the **structural chassis and mechanical transmission layout** in **AutoCAD and SolidWorks** for a multi-purpose utility vehicle prototype.

WORK EXPERIENCE

Feb 2024 – Jun 2024 **Computer-Aided Design Engineer** | *Chakra Technologies, Navi Mumbai, India*

- Reduced **design iterations before prototyping** by validating components with **FEM structural analysis in ANSYS Workbench**.
- Delivered **production-ready components and assemblies** by modeling them in **SolidWorks** with complete **drawings, BOMs and technical documentation**.
- Supported **production planning** by handing over complete **manufacturing documentation packages** to engineering and production teams.

Jul 2023 – Jan 2024 **Mechanical Design Engineer** | *LCT Pvt. Ltd., Navi Mumbai, India*

- Improved **test equipment reliability under 2–10 t loads** by introducing **5 design modifications** based on engineering evaluation.
- Cut **workshop preparation time by approx. 2 weeks** by coordinating with manufacturing during **assembly and commissioning**.
- Translated design requirements into **production-ready CAD assemblies and drawings** for industrial testing equipment.

Sep 2022 – Nov 2022 **Student Assistant** | *Disha Computer Institute of Technology, Navi Mumbai, India*

- Contributed to a **92 % first-attempt certification pass rate** by teaching **CAD, technical drawing and GD&T** practices.
- Improved **documentation quality** by reviewing **40+ CAD models and drawings** for inconsistencies.

SKILLS

Simulation: MATLAB, Simulink, Simscape Fluids, Model Predictive Control (MPC), Kalman Filter

EV Systems: Powertrain Modeling, Battery Thermal Management, SOC Estimation, Energy Management

Programming & Computer Vision: Python, OpenCV, NumPy, YOLOv8 (Ultralytics)

CAD & FEA: SolidWorks, CATIA V5, AutoCAD, ANSYS Workbench, GD&T

Testing & Validation: Test Equipment Commissioning, Prototype Field Testing, Oscilloscope, Multimeter

Documentation: Engineering Drawings, BOM Management, Technical Documentation

Tools: MS Office (Excel, PowerPoint), LaTeX, GitHub

CERTIFICATIONS

AI-Assisted Code Modernization with IBM *(IBM SkillsBuild Bobathon & ProoV evaluation)*: Passed with **96**/**100** by **diagnosing and fixing bugs in a legacy Python service** of a car company using an **AI coding agent**, **rejecting an unsafe AI-made change** and **verifying the fixes with unit tests**.

LANGUAGES & INTERESTS

Languages: German (Intermediate – B1), English (Fluent – B2/C1), Japanese (Beginner – N5), Hindi (Fluent – C1), Marathi (Native – C2)

Interests: Football *(FIFA World Cup & Champions League)*, Drawing B&W Sketches *(Human Portraits, Movies & Anime Characters)*, Photography *(Human Portraits, Landscapes, Wildlife)*, PC Gaming *(Story based)*

Weingarten, November 5, 2026



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